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K J SOMAIYA INSTITUTE OF TECHNOLOGY, SION, MUMBAI

DEPARTMENT OF COMPUTER ENGINEERING

AgriSense: AI-Driven Crop Type, Phenology, Moisture Stress Detection and Irrigation Advisory

TY Mini Project Report

**Submitted in partial fulfillment of the requirements
of the Subject Project Based Learning: Mini Project Lab-I (Sem V)**

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CERTIFICATE

This is to certify that the project entitled “AgriSense: AI-Driven Crop Type, Phenology, Moisture Stress Detection and Irrigation Advisory” is bonafide work of [Student Name 1], [Student Name 2], [Student Name 3], and [Student Name 4], submitted as a Third Year Semester V Mini Project in the subject of Project Based Learning: Mini Project Lab-I, Computer Engineering, for the academic year 2026-27.

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ABSTRACT

AgriSense is a remote-sensing and artificial-intelligence decision-support platform developed for crop monitoring and irrigation advisory over Karnal District, Haryana. The project integrates optical and microwave satellite observations, weather reanalysis, static soil properties, machine-learning models, a FAO-style crop-water-balance engine, and a role-specific web application. The current implementation uses real Sentinel-2-derived spectral indices and Sentinel-1 VV/VH backscatter from ten approximately 16-day satellite periods during the Kharif 2025 season. Sampling at 4,311 fixed Karnal locations produced a 43,110-row temporal table. Optical coverage varies substantially because of cloud contamination, including two dates with no valid optical observations, while Sentinel-1 SAR coverage remains 100% over all ten satellite periods. This characteristic motivated the explicit fusion of optical and radar data and independent fresh/carried observation tracking in the application.

A second data stream consists of nineteen real 8-day weather GeoTIFFs from 1 June to 23 October 2025 containing rainfall, air temperature, dew point, wind speed, surface pressure, solar radiation, potential evaporation, and validity information. A 24-band static soil stack represents texture, bulk density, organic carbon, and field capacity at six depth levels. These sources are combined through a 19-step 8-day master timeline: weather updates every eight days, while the latest valid optical and SAR observations are retained independently without inventing missing satellite images.

Four analytical engines are implemented. The Crop Engine compares Random Forest, XGBoost and RBF-SVM; the Phenology Engine compares Random Forest, XGBoost and a sequence-based LSTM; the Stress Engine compares Logistic Regression, Random Forest and XGBoost; and the Irrigation Engine applies a transparent FAO-56-style reference evapotranspiration and crop-coefficient water balance followed by priority rules. The LSTM operates on three-step sequences of NDVI, NDMI, NDWI, EVI, SAVI, VV, VH, the corrected VV-VH dB difference, observation-validity masks, and season day. The application exposes a 19-step seasonal timeline, live model inference, map layers, confusion matrices, and role-specific interfaces for agriculture officers, irrigation officers, Water User Associations, farmers, administrators, GIS analysts, and policy/research users.

The current model evaluation is explicitly treated as prototype evaluation because crop, stress, and phenology targets are derived or pseudo labels rather than independent field-survey ground truth. Therefore, the present work demonstrates a reproducible, app-ready multi-sensor architecture and realistic seasonal data integration rather than claiming fully validated agronomic accuracy. The report documents the literature basis, requirements, architecture, UML design, methodology, implementation, results, limitations, and future validation roadmap for AgriSense.

CONTENTS

TITLE	Page no.
LIST OF FIGURES	viii
LIST OF TABLES	x
LIST OF ABBREVIATION	xi
Chapter No.	TITLE / SECTION
1	INTRODUCTION
	1.1 Problem Definition 1
	1.2 Aim and Objective 1
	1.3 Organization of the Report 2
2	REVIEW OF LITERATURE
	2.1 Introduction 3
	2.2 Literature Survey 4
	2.3 Limitations of Existing Systems 8
3	REQUIREMENT SPECIFICATION
	3.1 Introduction 9
	3.2 Functional Requirements 10
	3.3 Non-Functional Requirements 11
	3.4 Hardware Requirements 12
	3.5 Software Requirements 12

CONTENTS (CONTINUED)

Chapter No.	TITLE / SECTION
4	PROJECT ANALYSIS & DESIGN
	4.1 Introduction 13
	4.2 System Architecture 14
	4.3 System Flow Diagram 15
	4.4 UML Diagrams 16
	4.5 Project Timeline 20
5	METHODOLOGY
	5.1 Introduction 21
	5.2 Proposed Methodology 21
	5.3 Data and API Schema 24
6	PROJECT IMPLEMENTATION AND RESULT
	6.1 Introduction 25
	6.2 System Implementation 25
	6.3 Performance Evaluation 29
7	CONCLUSION & FUTURE SCOPE
	7.1 Conclusion 32
	7.2 Limitations 32
	7.3 Future Scope 32
8	REFERENCES 34

LIST OF FIGURES

Figure No.	Title	Page No.
4.1	System Architecture of AgriSense	14
4.2	System Flow Diagram	15
4.3	UML Use Case Diagram	16
4.4	UML Activity Diagram	17
4.5	UML Sequence Diagram	18
4.6	UML Class Diagram	19
4.7	Project Development Timeline	20
5.1	AI and Data-Fusion Pipeline	22

LIST OF FIGURES (CONTINUED)

Figure No.	Title	Page No.
6.1	Karnal Kharif 2025 Satellite Observation Coverage	26
6.2	8-Day Weather Timeline	26
6.3	Crop Engine - Random Forest Confusion Matrix	30
6.4	Phenology Engine - LSTM Confusion Matrix	30
6.5	Stress Engine - XGBoost Confusion Matrix	31
6.6	Application Screenshot - Agriculture Officer Dashboard	27
6.7	Application Screenshot - Irrigation Officer Dashboard	27
6.8	Application Screenshot - Farmer View	27
6.9	Application Screenshot - AI Model Evaluation	28

LIST OF TABLES

Table No.	Title	Page No.
2.1	Summary of Literature Survey - Related Works	4
2.2	Limitations Identified from Existing Systems	8
3.1	Functional Requirements	10
3.2	Non-Functional Requirements	11
3.3	Hardware Requirements	12
3.4	Software Requirements	12
4.1	Project Development Timeline Summary	20
5.1	AgriSense Data Sources and Native Resolution	21
5.2	Real Satellite Temporal Table Schema	24
5.3	AI Engine Configuration	24
5.4	Data and API Endpoint Schema	24
6.1	Satellite Temporal Coverage by Period	25
6.2	Prototype Model Performance Summary	29
6.3	Result Interpretation and Validation Status	31

LIST OF ABBREVIATIONS

Sr. No.	Abbreviation	Description
1	AI	Artificial Intelligence
2	API	Application Programming Interface
3	AOI	Area of Interest
4	CRS	Coordinate Reference System
5	ET ₀	Reference Evapotranspiration
6	ETc	Crop Evapotranspiration
7	EVI	Enhanced Vegetation Index
8	FAO	Food and Agriculture Organization
9	GEE	Google Earth Engine
10	Kc	Crop Coefficient
11	LSTM	Long Short-Term Memory
12	ML	Machine Learning
13	MSI	Multispectral Instrument
14	NDMI	Normalized Difference Moisture Index
15	NDVI	Normalized Difference Vegetation Index
16	NDWI	Normalized Difference Water Index
17	PEV	Potential Evaporation
18	RF	Random Forest
19	SAR	Synthetic Aperture Radar
20	SAVI	Soil Adjusted Vegetation Index
21	S1	Sentinel-1
22	S2	Sentinel-2
23	SVM	Support Vector Machine
24	UI	User Interface
25	WUA	Water User Association
26	XGB	Extreme Gradient Boosting / XGBoost

CHAPTER 1

INTRODUCTION

1.1 Problem Definition

Agriculture in Karnal requires timely information on crop type, seasonal growth stage, moisture stress and irrigation need. Field scouting is accurate but difficult to repeat frequently over large areas. Satellite remote sensing offers scalable monitoring, but optical and microwave sensors observe different aspects of the crop and have different limitations.

Sentinel-2 provides multispectral vegetation and moisture-sensitive information, while Sentinel-1 C-band SAR provides cloud-independent VV/VH observations. During the Kharif monsoon, cloud cover can remove most or all usable optical observations on some dates; therefore AgriSense fuses optical, SAR, weather and soil information rather than depending on a single sensor.

The implemented Karnal dataset contains 4,311 fixed locations across ten real satellite dates (43,110 temporal rows). SAR is valid across all ten dates, whereas optical coverage ranges from 0% to 100%. The system must preserve this missingness honestly, track the age of carried observations, and never fabricate an 8-day satellite image.

1.2 Aim and Objective

Aim

To design and implement AgriSense, an integrated satellite remote-sensing and AI decision-support platform for Karnal District that combines crop-type identification, phenology-aware moisture-stress monitoring and 8-day irrigation advisory using optical, microwave, weather and soil data.

Objectives

1. Develop a reproducible data pipeline for Sentinel-2 optical indices and Sentinel-1 VV/VH SAR time series over Karnal.
2. Create a real temporal feature table and preserve optical/SAR validity rather than replacing missing observations with fabricated satellite data.
3. Compare Random Forest, XGBoost and SVM models for the Crop Engine.
4. Compare Random Forest and XGBoost baselines with an LSTM temporal model for the Phenology Engine.
5. Compare Logistic Regression, Random Forest and XGBoost for moisture-stress classification.
6. Integrate 19 real 8-day weather periods and a 24-band soil stack with the satellite timeline.
7. Implement a FAO-56-style crop-water-balance and irrigation-priority rule engine.
8. Develop a role-specific web application with seasonal playback, map layers, field inspection, model evaluation and persistent Python inference.

1.3 Organization of the Report

The remainder of this report is organized as follows:

Chapter	Content Description
Chapter 2	Review of Literature - reviews crop classification, SAR-optical fusion, field-scale phenology, moisture-stress monitoring, evapotranspiration and irrigation-scheduling research.
Chapter 3	Requirement Specification - defines functional, non-functional, hardware and software requirements for the present prototype.
Chapter 4	Project Analysis & Design - presents the system architecture, system flow, UML use case/activity/sequence/class diagrams and project timeline.
Chapter 5	Methodology - explains data acquisition, temporal fusion, feature engineering, machine-learning engines, LSTM sequence preparation, water-balance logic and API/data schemas.
Chapter 6	Project Implementation and Result - documents the real Karnal dataset, app implementation, live inference architecture, observation coverage and prototype model evaluation.
Chapter 7	Conclusion & Future Scope - summarizes outcomes, identifies present limitations and defines the validation and production roadmap.
Chapter 8	References - lists the scientific, methodological and technology sources cited throughout the report.

The project deliberately distinguishes between measured observations, algorithmically derived variables and prototype AI outputs. This distinction is carried throughout the report because the current crop, phenology and stress labels are not yet independent field-survey ground truth.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Introduction

Remote-sensing-based agricultural monitoring has progressed from single-date crop mapping to dense multi-sensor time-series analysis. Sentinel-2 offers fine-resolution spectral information suited to vegetation and water-sensitive indices, while Sentinel-1 SAR provides cloud-independent temporal measurements. Modern studies increasingly combine these sensors with machine learning, phenological features, environmental data and cloud platforms such as Google Earth Engine. This literature review concentrates on research that directly informs the AgriSense design: crop type classification, phenology retrieval, SAR-optical fusion, moisture-stress detection, irrigation scheduling and scalable geospatial processing.

The relevance to Karnal is particularly strong because the monsoon season creates severe optical-data gaps. ESA reports a five-day revisit for the two-satellite Sentinel-2 constellation and 13 MSI spectral bands at 10-60 m resolution, but revisit frequency does not guarantee cloud-free observations [1]. Sentinel-1 time series have therefore become important for crop mapping and seasonal monitoring in cloudy agricultural regions. Haryana-specific studies have already demonstrated the value of Sentinel-1 for regional crop classification, while recent work around Karnal has reported strong Random Forest and SVM performance using dual-polarization SAR vegetation indices [5][6].

AgriSense extends these directions by combining a real Kharif 2025 Sentinel-1/Sentinel-2 temporal table with a separate 8-day weather timeline, soil context, three ML engines, a sequence-based phenology model and a transparent irrigation water-balance engine. The literature below is therefore evaluated not only for reported accuracy, but also for assumptions about ground truth, temporal density, cloud robustness, transferability and suitability for operational decision support.

2.2 Literature Survey

Crop Classification and Sentinel-1/Sentinel-2 Fusion

Mishra et al. investigated rainy-season crop classification near Karnal using time-series Sentinel-1 C-band SAR and dual-polarization vegetation indices derived from VV and VH. The study emphasized the ability of radar to penetrate thick cloud cover and reported classification accuracies of 93.77% for Random Forest and 93.50% for SVM [5]. This work is geographically significant for AgriSense and supports the decision to retain SAR as a continuous seasonal signal even when optical data are unavailable.

Ahlawat et al. mapped the Rabi cropping pattern of Hisar District, Haryana using multi-date Sentinel-1 observations and a multi-data approach [6]. Wheat, mustard, sugarcane and other crops were separated from multi-temporal radar behaviour. Although the study used a different season and classification procedure, it demonstrates the feasibility of regional crop mapping in Haryana from repeated SAR observations.

Cai, Lin and Zhang proposed an object-based Random Forest workflow for paddy rice mapping using time-series Sentinel-1 and Sentinel-2 data [7]. Their method combined optical NDVI, phenological parameters and SAR backscatter, achieving overall accuracy above 95% in their study area. The result reinforces the importance of temporal and multi-modal features rather than relying on a single image.

Blickensdörfer et al. combined Sentinel-1, Sentinel-2 and Landsat 8 time series for national-scale crop-type and crop-sequence mapping in Germany [8]. Their workflow used a Random Forest classifier and environmental context across 24 agricultural land-cover classes. The study is important to AgriSense because it shows that optical, radar and environmental features can jointly improve robustness under inter-annual variability.

Recent work on smallholder agriculture in Central and South Asia reported that Sentinel-1/2 fusion improves crop classification and that temporal aggregation and feature selection are especially important when fields are fragmented and labels are limited [14]. This aligns with the Karnal design, where real temporal signatures are prioritized and model evaluation must explicitly account for class imbalance and incomplete labels.

Crop Phenology from Optical and SAR Time Series

Woźniak et al. derived multi-temporal phenological indices from Sentinel-1 image time series for country-wide crop classification in Poland [9]. Their radar-based approach achieved regional overall accuracies of approximately 86-90%, and the proposed phenological indices improved individual crop F1 scores. The study shows that seasonal crop structure is encoded in SAR temporal behaviour even when optical observations are unavailable.

Meroni et al. compared crop-specific land-surface phenology derived from Sentinel-1 cross-polarization behaviour and Sentinel-2 NDVI for major European crops [10]. Their analysis found that Sentinel-1 and Sentinel-2 provide relevant and sometimes complementary phenology information. This is directly reflected in AgriSense's independent optical/SAR freshness logic: a SAR observation may be fresh even while NDVI must be carried from an earlier cloud-free optical date.

Liao et al. proposed near-real-time within-field phenology detection and forecasting for winter wheat and corn using Sentinel-2 time series [11]. Their framework estimated detailed BBCH stages and reported stage-date errors generally below four days for the studied crops. The work demonstrates that continuous growth-stage estimation requires temporal shape information rather than a static spectral snapshot.

Field-scale phenology research also highlights a central operational limitation: high spatial resolution is valuable for individual fields, but cloud-free temporal density may be insufficient. Fusion of high-spatial-resolution and high-temporal-resolution data has therefore been used to improve field-scale phenometrics [12]. AgriSense addresses the same issue conservatively by retaining the actual acquisition dates and marking carried-forward observations rather than synthesizing additional satellite images.

Moisture Stress, Evapotranspiration and Irrigation Scheduling

Moisture stress is difficult to infer from a single spectral index because canopy condition, crop stage, soil background, weather and management influence the signal. Veysi et al. evaluated a satellite-based crop water stress index for irrigation scheduling in sugarcane and validated satellite-derived stress against field measurements of canopy temperature and vegetation water content [13]. Their results demonstrate the value of satellite stress information for irrigation scheduling, but also emphasize the need for independent in-situ validation.

For AgriSense, optical moisture indicators such as NDMI and NDWI are combined with Sentinel-1 backscatter and phenological context. This strategy is motivated by the complementary sensitivity of optical reflectance and microwave backscatter to canopy and surface conditions. However, the present stress labels are derived/pseudo labels and are strongly imbalanced; therefore high prototype accuracy is not treated as independent evidence of agronomic stress-detection accuracy.

The irrigation engine is based on the FAO-56 crop evapotranspiration framework of Allen et al. [3]. FAO-56 computes reference evapotranspiration ET_0 from meteorological variables using the Penman-Monteith formulation and obtains crop evapotranspiration from $ET_c = K_c \times ET_0$, where K_c changes with crop and growth stage. AgriSense follows this transparent structure, combines it with effective rainfall, and converts the remaining deficit into a prototype irrigation-priority rule.

ERA5-Land provides globally consistent, hourly land variables at approximately 9 km native resolution and is widely used in hydrological and environmental applications [2]. The AgriSense weather exports are aggregated into 8-day rasters containing rainfall, mean/max/min temperature, dew point, wind, surface pressure, solar radiation and potential evaporation. These variables permit an 8-day advisory cadence even when satellite observations occur less frequently.

Machine Learning and Scalable Geospatial Processing

Random Forest is widely used in crop mapping because it handles nonlinear feature interactions, mixed feature scales and relatively small training datasets while providing robust performance [15]. XGBoost implements scalable gradient-boosted decision trees and is useful when complex decision boundaries can be learned from tabular remote-sensing features [16]. Support Vector Machines remain competitive in remote-sensing classification, particularly when class boundaries are well separated in feature space. AgriSense compares all three for crop classification rather than assuming one classifier a priori.

For temporal phenology, Long Short-Term Memory networks are designed to learn dependencies across sequential observations through gated recurrent units [17]. The AgriSense LSTM therefore uses ordered three-step sequences containing optical indices, VV/VH, validity masks and season day. The temporal split is performed by field identifier, preventing observations from the same location appearing in both training and test sets.

Google Earth Engine has become a common platform for large-scale geospatial preprocessing because it co-locates remote-sensing archives with cloud computation [4]. AgriSense used GEE to prepare optical, SAR, weather and soil exports, but the final prototype is deliberately capable of operating from downloaded local GeoTIFFs and CSV files. This was necessary because free-tier computational quotas limited repeated extraction of the complete corrected 8-day satellite stack.

Table 2.1: Summary of Literature Survey - Related Works

Study	Problem	Method / Data	Key Finding	Relevance / Limitation
Mishra et al., 2022 [5]	Karnal-region rainy-season crop classification	Sentinel-1 VV/VH + dual-pol indices; RF/SVM	RF 93.77%, SVM 93.50%; strong cloud robustness	Uses SAR-focused crop labels; does not integrate irrigation decision support
Ahlawat et al., 2020 [6]	Regional crop mapping in Hisar, Haryana	Multi-date Sentinel-1	Mapped wheat, mustard, sugarcane and other crops	Different season and methodology; limited multi-sensor fusion
Cai et al., 2019 [7]	Paddy rice mapping	Sentinel-1 + Sentinel-2 + phenology + RF	Overall accuracy >95%	Specific to rice and study region
Blickensdörfer et al., 2022 [8]	Crop types and crop sequences	S1 + S2 + Landsat + RF	Large-area multi-year crop mapping	Requires extensive reference data
Woźniak et al., 2022 [9]	Phenological indices for crop classification	Sentinel-1 time series	Approx. 86-90% regional OA; phenology improves F1	Radar phenology not directly equivalent to agronomic stages
Meroni et al., 2021 [10]	Crop land-surface phenology	S1 cross-ratio + S2 NDVI	Optical/SAR phenology can be complementary	Agreement varies by crop and stage
Liao et al., 2023 [11]	Near-real-time crop phenology	Sentinel-2 time series + shape models	Detailed BBCH stage estimation	Cloud-free S2 temporal density remains a constraint
Veysi et al., 2017 [13]	Water stress and irrigation scheduling	Landsat thermal CWSI + field validation	Demonstrated operational stress classes	Relies on thermal data and in-situ validation
Raab et al., 2026 [14]	Smallholder crop classification	Sentinel-1/2 fusion + RF	Fusion and temporal aggregation improve mapping	Ground truth and transferability remain central
Allen et al., 1998 [3]	Crop water requirements	FAO-56 Penman-Monteith + Kc	Standard transparent ET framework	Requires locally appropriate crop coefficients and weather

2.3 Limitations of Existing Systems

The reviewed literature establishes the technical feasibility of multi-temporal crop classification, phenology monitoring and satellite-supported irrigation management, but several limitations remain when these methods are transferred into a single operational application for an Indian command-area context.

Table 2.2: Limitations Identified from Existing Systems

Limitation	Effect	AgriSense Design Response
Optical cloud dependence	Sentinel-2 time series can lose critical monsoon observations.	Use Sentinel-1 as an independent all-weather temporal stream and expose validity/age.
Ground-truth dependence	High reported accuracy often relies on extensive parcel labels or field surveys.	Current AgriSense results are labelled prototype until independent Karnal field truth is collected.
Phenology interpretation	Remote-sensing phenometrics do not always correspond directly to agronomic growth stages.	Use temporal LSTM plus explicit pseudo-label warning; future work requires observed stage labels.
Single-task systems	Many studies focus only on crop mapping, phenology or water stress.	AgriSense integrates crop, stage, stress, weather, soil and advisory in one timeline.
Temporal mismatch	Different sensors and weather products operate at different cadences.	Use a 19-step 8-day master calendar while retaining native satellite dates and observation age.
Irrigation validation	Remote-sensing stress or ET estimates are often not tested against actual irrigation events.	Current advisory is rule-based prototype; future validation must use soil-moisture/irrigation records.
Transferability	Models trained in one region or season may degrade elsewhere.	Use grouped spatial/field evaluation and future multi-year validation.
Operational transparency	Black-box outputs can be difficult for officers and farmers to interpret.	Retain real/derived/prototype labels, source dates, confidence and role-specific views.

The main research gap addressed by the present prototype is therefore the engineering of an end-to-end, provenance-aware seasonal system rather than a claim that all agronomic prediction problems have already been solved. AgriSense treats data availability, temporal consistency and validation status as first-class system properties.

CHAPTER 3

REQUIREMENT SPECIFICATION

3.1 Introduction

AgriSense is implemented as a remote-sensing data pipeline, machine-learning inference service, irrigation decision engine and browser-based geospatial application. The requirement specification defines what the system must do, how reliably it should operate, the computing environment required for development, and the software technologies used by the current prototype.

The system is designed for multiple stakeholder roles. Agriculture Officers require broad regional crop-condition and model-evaluation views; Irrigation Officers and WUAs require water-deficit and advisory layers; Farmers require a simplified view restricted to assigned plots; GIS/RS Analysts require temporal and QA information; and Administrators or Researchers require access to pipeline and evaluation outputs.

Because the prototype uses real observations mixed with derived and prototype components, a critical requirement is provenance visibility. Every period should indicate whether optical and SAR data are fresh, carried from an earlier valid observation, or unavailable. The application must never imply that an interpolated or carried observation is a newly acquired satellite image.

3.2 Functional Requirements

Table 3.1: Functional Requirements

ID	Requirement	Description
FR-01	Load Karnal satellite temporal data	System shall load the 4,311-location × 10-date real satellite temporal table.
FR-02	Support variable-band periods	System shall handle full 14-band and SAR-only 3-band old GEE exports without failure.
FR-03	Track sensor validity	System shall retain optical_valid and sar_valid masks and source dates.
FR-04	Master seasonal timeline	System shall expose 19 8-day periods from 1 June to 23 October 2025.
FR-05	Fresh/carried state	System shall carry optical and SAR independently and display observation age.
FR-06	Crop Engine	System shall provide crop predictions using the selected Random Forest deployment model.
FR-07	Phenology Engine	System shall provide stage predictions using the LSTM temporal model.
FR-08	Stress Engine	System shall provide moisture-stress predictions using the selected XGBoost model.
FR-09	Irrigation Engine	System shall estimate ET ₀ , ET _c , effective rainfall, water deficit and irrigation priority.
FR-10	Role-specific UI	System shall show different map layers and controls for Officer, Irrigation, WUA, Farmer and Policy roles.
FR-11	Field Inspector	System shall display current satellite source dates, indices, weather, water balance and model outputs.
FR-12	Model Evaluation	System shall display stored accuracy, macro metrics and confusion matrices.
FR-13	Persistent inference	System shall load serialized model artifacts once and reuse them across API requests.
FR-14	Export	System shall allow field-level output export and retain audit metadata.

3.3 Non-Functional Requirements

Table 3.2: Non-Functional Requirements

ID	Attribute	Requirement
NFR-01	Transparency	Real observations, derived variables, carried states and prototype outputs shall be visibly distinguished.
NFR-02	Reproducibility	Data-generation and temporal-fusion steps shall be scripted and deterministic.
NFR-03	Scalability	The architecture shall separate preprocessing, period data serving and model inference so components can scale independently.
NFR-04	Usability	Stakeholders shall see only the information relevant to their role in the demo interface.
NFR-05	Maintainability	Frontend, Node API and Python inference shall remain modular and independently replaceable.
NFR-06	Performance	Serialized ML artifacts shall be loaded once per inference-service process, avoiding model reload on every request.
NFR-07	Robustness	Missing optical values shall not terminate the application; SAR-only periods shall remain usable.
NFR-08	Data Integrity	VV/VH processing shall use the correct dB-domain difference VV - VH rather than the invalid direct quotient from the first script.
NFR-09	Security	Model artifacts shall not be served as public static files. Production deployment will require server-side authentication/authorization.
NFR-10	Scientific Caution	Prototype confusion matrices shall not be presented as independent field validation.

3.4 Hardware Requirements

Table 3.3: Hardware Requirements

Component	Specification	Purpose
Development CPU	4-core 64-bit processor or better	Python raster processing, model inference and Node server
RAM	8 GB minimum; 16 GB recommended	GeoTIFF sampling, Pandas temporal tables, TensorFlow LSTM
Storage	20 GB free SSD minimum	Satellite TIFFs, temporal CSV/Parquet, model artifacts and app data
GPU	Optional NVIDIA CUDA-capable GPU	Speeds LSTM training; not required for inference/demo
Client	Modern desktop/laptop browser	Leaflet map, Chart.js charts and role-specific dashboard
Network	Stable internet during data acquisition	GEE/Copernicus access and package installation; local demo can run offline except basemap tiles

3.5 Software Requirements

Table 3.4: Software Requirements

Layer	Technology	Role
Geospatial preprocessing	Google Earth Engine; Rasterio; PyProj	GEE export and local GeoTIFF sampling/reprojection
Data processing	Python 3.x; Pandas; NumPy; PyArrow	Temporal tables, feature engineering and Parquet/CSV
ML - tabular	scikit-learn; XGBoost	RF, SVM, Logistic Regression and XGBoost
ML - sequence	TensorFlow / Keras	Phenology LSTM training and inference
Backend	Node.js + Express	Static app server, period API and inference proxy
Inference service	FastAPI + Uvicorn	Persistent Python model loading and prediction endpoints
Frontend	HTML/CSS/JavaScript; Leaflet; Chart.js	Maps, charts, seasonal timeline and role-specific UI
Version control	Git / GitHub	Source-code management and reproducible development

CHAPTER 4

PROJECT ANALYSIS & DESIGN

4.1 Introduction

The AgriSense design separates data preparation, temporal fusion, analytical engines, service APIs and the stakeholder-facing web interface. The architecture must support two different temporal cadences: real satellite observations at approximately 16-day dates from the existing download and real 8-day weather periods from the corrected GEE weather export. The master calendar therefore updates every eight days while preserving the true source date and age of optical and SAR observations.

A second design concern is validation provenance. The application contains real observed features, derived physical variables, pseudo labels used for prototype machine learning, and decision rules. These categories are kept separate in the UI and report. This avoids a common failure mode in prototype agricultural systems where derived labels are inadvertently described as independently validated truth.

4.2 System Architecture

Figure 4.1 shows the end-to-end architecture. Satellite observations are transformed into a real temporal table; the weather and soil streams are fused through the 19-step master timeline; analytical engines generate crop, stage, stress and irrigation outputs; a persistent FastAPI process holds selected serialized models in memory; Node/Express proxies the inference and period-data APIs; and the browser presents role-specific maps and analytics.

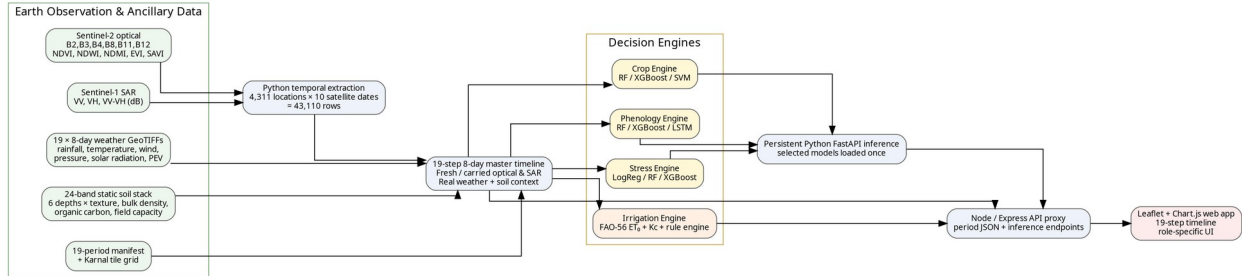


Figure 4.1: System Architecture of AgriSense

The architecture intentionally keeps large preprocessing tasks outside the live request path. Period JSON files and indexed temporal data are prepared in advance, while model inference loads only the required field history for a prediction. This design reduces repeated disk/model loading and supports future migration to PostGIS, object storage and tiled raster services without replacing the user interface.

4.3 System Flow Diagram

The operational flow begins with real data acquisition and preprocessing, followed by temporal-table generation and master-period construction. Model inference and irrigation computation are then executed for the selected field and period. The final stage is role-aware visualization and export.

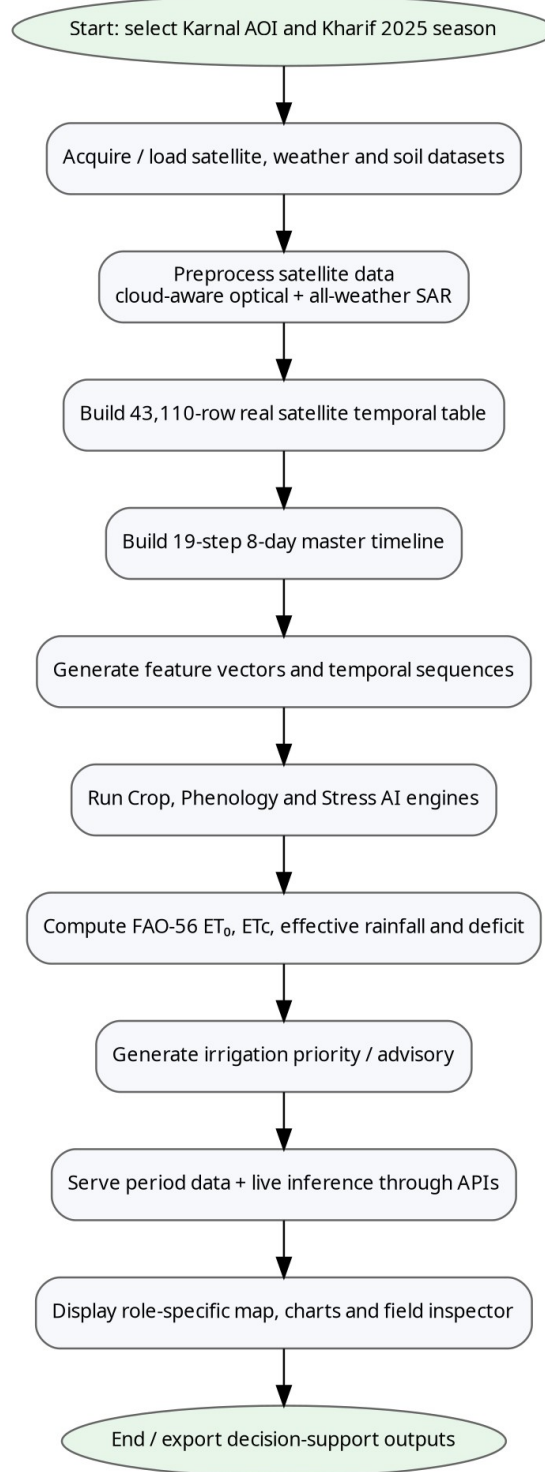


Figure 4.2: System Flow Diagram

4.4 UML Diagrams

4.4.1 Use Case Diagram

The use case diagram defines the interaction of seven stakeholder roles with the AgriSense platform. Farmers are intentionally restricted to assigned demo plots, whereas officers and analysts receive broader regional and technical views.

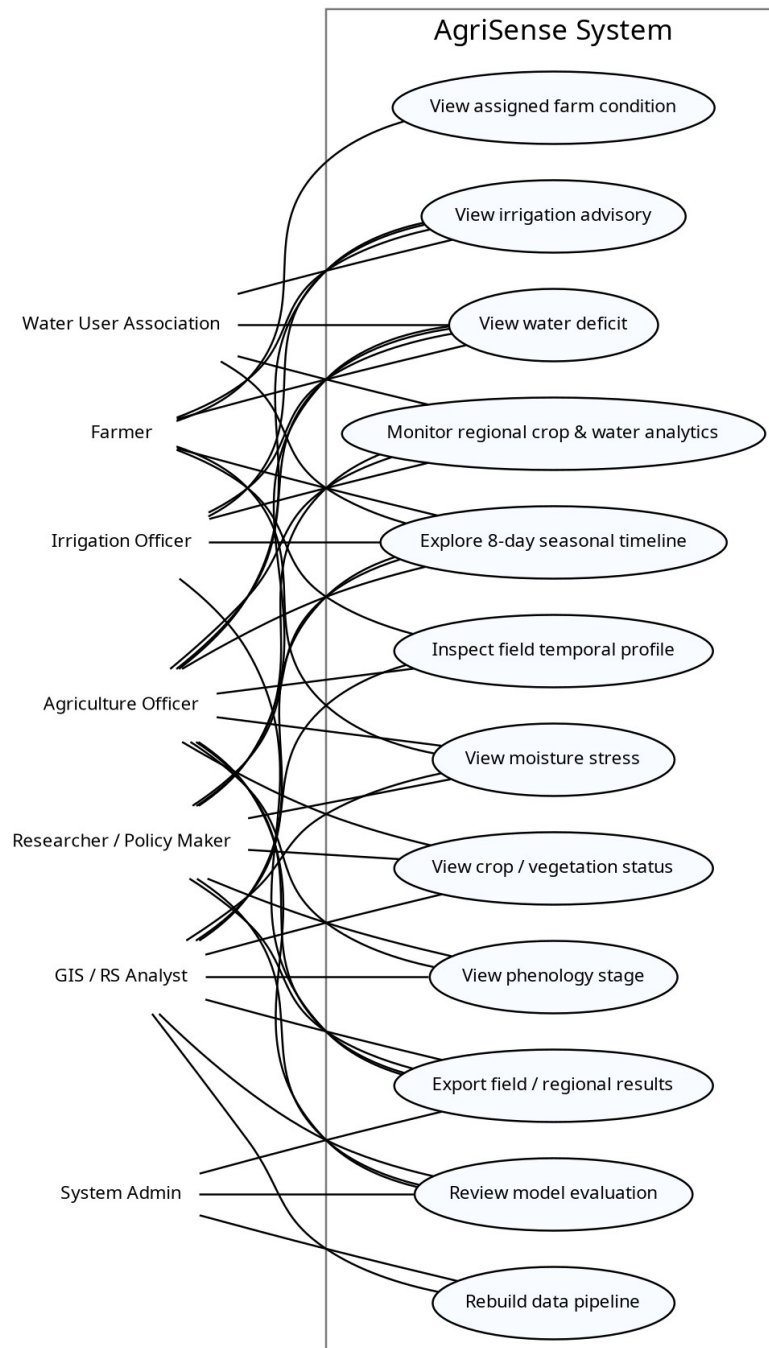


Figure 4.3: UML Use Case Diagram

4.4.2 Activity Diagram

The activity model captures the processing path for a selected master period. The key branch is satellite availability: current measurements are used when fresh, while optical and SAR streams are carried independently from their most recent valid source date when necessary.

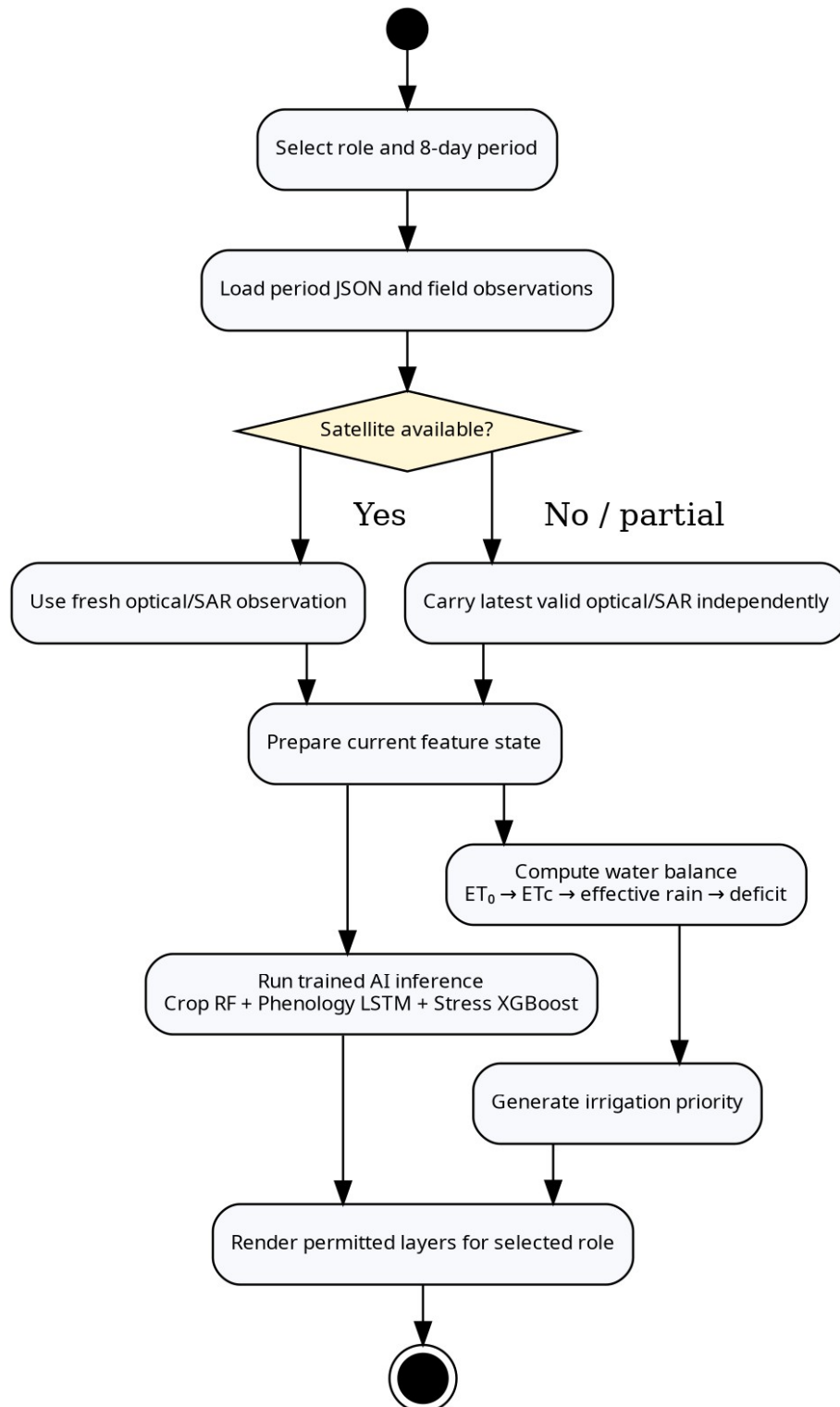


Figure 4.4: UML Activity Diagram

4.4.3 Sequence Diagram

The sequence diagram represents a live field-assessment request. The browser first loads the selected period from Node/Express, then requests model inference. Node proxies the request to the persistent Python service, which calls the in-memory Crop RF, Phenology LSTM and Stress XGBoost models and returns probabilities and labels.

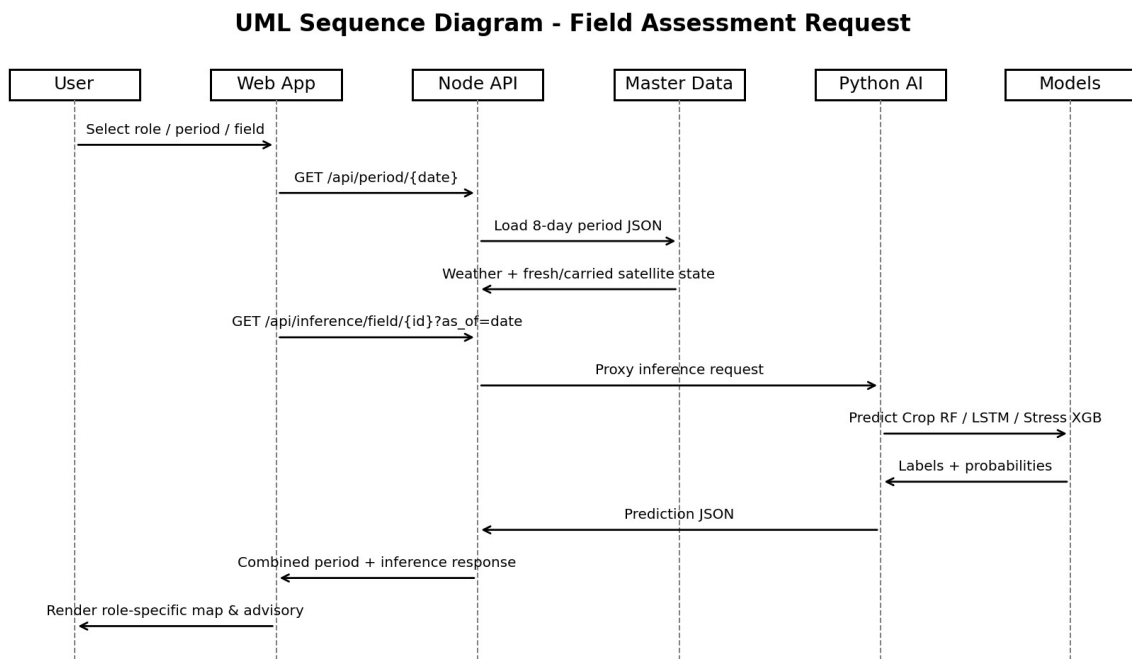


Figure 4.5: UML Sequence Diagram

4.4.4 Class Diagram

The class diagram presents the principal domain entities used by the implementation: a field/observation unit, master period, satellite observation, weather state, model prediction, irrigation advisory and user role. The implementation is file- and JSON-oriented rather than database-backed, but these classes provide a direct path to a future relational or spatial database schema.

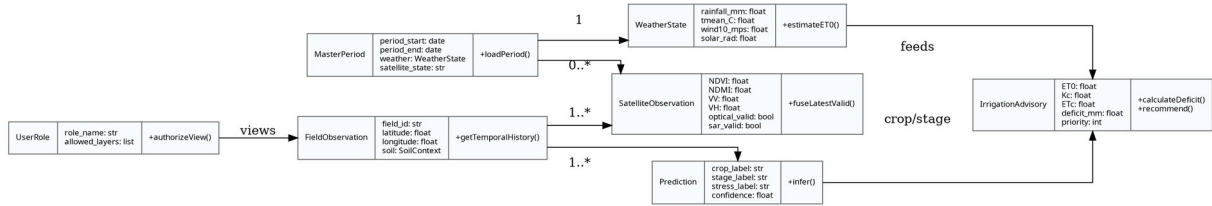


Figure 4.6: UML Class Diagram

4.5 Project Timeline

Table 4.1: Project Development Timeline Summary

Phase	Activity	Duration	Week No.
Phase 1	Problem definition and literature review	1 week	Weeks 1-2
Phase 2	GEE data design and Karnal satellite extraction	2 weeks	Weeks 2-4
Phase 3	Real temporal table generation and QA	1 week	Weeks 4-5
Phase 4	Crop / phenology / stress model experiments	2 weeks	Weeks 5-7
Phase 5	8-day weather and soil integration	2 weeks	Weeks 6-8
Phase 6	Web application and role-specific UI	2 weeks	Weeks 7-9
Phase 7	Persistent Python inference integration	2 weeks	Weeks 8-10
Phase 8	Testing, report preparation and presentation	2 weeks	Weeks 10-12

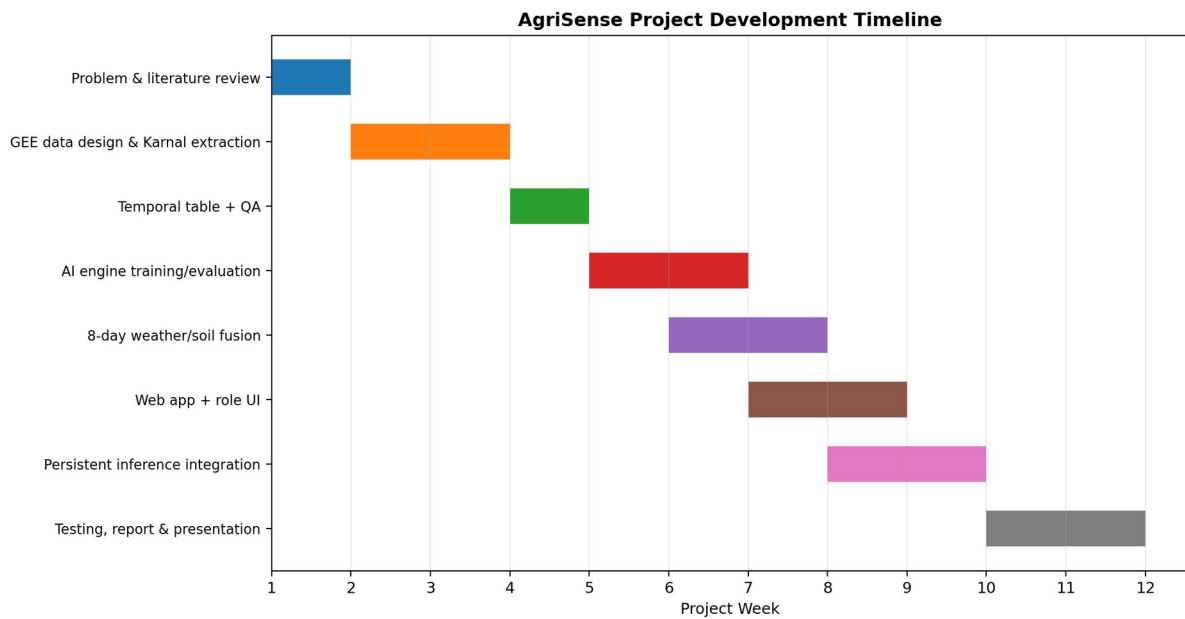


Figure 4.7: Project Development Timeline

CHAPTER 5

METHODOLOGY

5.1 Introduction

The AgriSense methodology is organized as a provenance-aware multi-source temporal pipeline. It begins with real satellite, weather and soil data, creates a consistent seasonal table, extracts tabular and sequence features, evaluates multiple machine-learning models, and connects the selected models to a persistent inference API. The irrigation engine remains physics/rule based so its assumptions are transparent.

5.2 Proposed Methodology

5.2.1 Study Area and Seasonal Window

Karnal District, Haryana is used as the pilot study area. The implemented demonstration focuses on the Kharif 2025 season. The old satellite download covers ten composite start dates: 1 June, 17 June, 3 July, 19 July, 4 August, 20 August, 5 September, 21 September, 7 October and 23 October 2025. A corrected weather export provides nineteen 8-day periods from 1 June to 23 October 2025.

5.2.2 Data Acquisition

Table 5.1: AgriSense Data Sources and Native Resolution

Source	Origin	Temporal / Spatial Context	Variables
Sentinel-2 optical	Old GEE export	~16-day dates; pixel features at 10/20 m source scale	B2, B3, B4, B8, B11, B12, NDVI, NDWI, NDMI, EVI, SAVI
Sentinel-1 SAR	Old GEE export	~16-day dates; C-band dual polarization	VV, VH, corrected VV-VH dB difference
Weather	New GEE export	19 × 8-day periods; 5.5 km raster	Rainfall, Tmean/Tmax/Tmin, dew point, wind, pressure, solar radiation, ERA5 PEV, valid flag
Soil	New static stack	250 m; six depth levels	Texture, bulk density, organic carbon, field capacity
Manifest / tile grid	New GEE export	19 periods / 24-tile export grid	Period dates and export metadata

5.2.3 Satellite Temporal Table Construction

The old download contains 110 satellite GeoTIFFs spanning ten dates. Eighty-eight TIFFs contain the full 14-band optical+SAR layout and twenty-two contain only the three SAR bands. The temporal builder detects GeoTIFF band descriptions and supports both layouts. For each of the 4,311 fixed Karnal locations, it samples whichever tile contains the coordinate and writes a canonical row with missing optical variables left as NaN when they are not observed.

The original first-script ratio feature is not used for modelling because Sentinel-1 GRD values are in dB. The corrected polarization difference is therefore computed as:

$$\text{VV_VH_ratio_dB} = \text{VV_dB} - \text{VH_dB}$$

The resulting table contains 43,110 rows (4,311 locations × 10 dates). Sensor-validity masks are retained as explicit features. EVI values outside the plausible range [-1, 1] are converted to missing values before LSTM preprocessing because unstable denominators can produce extreme EVI outliers.

5.2.4 19-Step 8-Day Master Timeline

The application does not resample or invent satellite imagery at 8-day intervals. Instead, each 8-day master period loads real weather for that interval and identifies the most recent valid optical and SAR

records on or before the period start. Optical and SAR are carried independently. Thus, a period can legitimately show 'OPT 16d OLD' and 'SAR FRESH'. The source date and age are shown in the user interface.

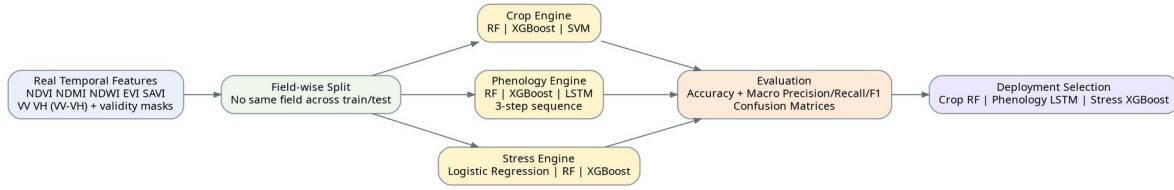


Figure 5.1: AI and Data-Fusion Pipeline

5.2.5 Crop Engine

The Crop Engine compares Random Forest, XGBoost and RBF-SVM using optical indices and corrected SAR features. The current labels are limited and derived, with the prototype classification task dominated by an Unclassified class and a smaller Paddy (Rice) class. Because of this label limitation, the confusion matrices demonstrate pipeline behaviour rather than a final multi-crop agronomic benchmark.

5.2.6 Phenology Engine

Two tabular baselines (Random Forest and XGBoost) are compared with a temporal LSTM. The corrected LSTM uses an 11-feature vector: NDVI, NDMI, NDWI, EVI, SAVI, VV, VH, VV-VH dB difference, optical_valid, sar_valid and season_day. Causal forward-fill is used only for previously observed optical values; no future back-fill is allowed. Missing values are imputed using a training-fitted median imputer, then standardized using a training-fitted scaler.

Three satellite steps form one LSTM input sequence. Fields are split by field_id into 3,017 training fields, 647 validation fields and 647 held-out test fields. Since independent stage labels are not yet available, presentation-only stage targets are derived from the real NDVI trajectory. The five output classes are Sowing/Planting, Vegetative, Flowering, Grain Filling and Maturity.

5.2.7 Moisture-Stress Engine

Logistic Regression, Random Forest and XGBoost are compared for moisture-stress classification. The input features use the same optical/SAR snapshot family as the Crop Engine. The current stress labels are highly imbalanced and derived, so macro metrics are reported alongside accuracy and the evaluation status is explicitly marked as prototype.

5.2.8 Irrigation Engine

The irrigation component uses a transparent FAO-style water balance rather than a black-box classifier. An 8-day reference evapotranspiration estimate is calculated from temperature, dew point, wind, surface pressure and solar radiation following the FAO-56 Penman-Monteith formulation [3]. Crop evapotranspiration is computed as $ET_c = K_c \times ET_0$. Effective rainfall is represented by a simple prototype fraction of period rainfall, and the irrigation deficit is calculated as the positive remainder after effective rainfall.

$$ET_c = K_c \times ET_0$$

$$Pe_{ff} = 0.80 \times \text{Rainfall}$$

$$\text{Deficit} = \max(0, ET_c - Pe_{ff})$$

The deficit is mapped to Low, Medium and High priority thresholds, with an escalation when the current NDMI-derived moisture signal is very low. This engine is intentionally labelled prototype until it is compared against observed irrigation events, root-zone soil moisture or trusted water-delivery records.

5.2.9 Persistent Model Inference

The deployment layer selects Random Forest for crop prediction, LSTM for temporal phenology and XGBoost for stress prediction. A FastAPI service loads the serialized model artifacts, encoders, LSTM imputer and scaler once during startup. The Node/Express server proxies browser requests to the Python service and blocks the `private_models` directory from public static access. This separates web delivery from Python ML execution and avoids reloading large model objects for every field click.

5.2.10 Role-Specific User Interface

The Agriculture Officer receives the complete crop, stage, stress, deficit, advisory, QA and model-evaluation interface. The Irrigation Officer and WUA views are limited to deficit and advisory functions. Farmer View displays only two assigned demo observation points and hides technical pipeline/model controls. Policy/Research views emphasize regional seasonal summaries. These restrictions are currently presentation-level controls; production use will require authenticated server-side authorization.

5.3 Data and API Schema

Table 5.2: Real Satellite Temporal Table Schema

Field	Type	Description
field_id	String	Stable observation / field identifier
period_start	Date	Real old-satellite composite start date
latitude, longitude	Float	WGS84 coordinate of sample location
NDVI, NDMI, NDWI, EVI, SAVI	Float	Optical vegetation/moisture indices; may be missing
VV, VH	Float dB	Sentinel-1 backscatter
VV_VH_ratio_dB	Float dB	Corrected polarization difference VV - VH
optical_valid	0/1	Whether valid optical data exist for this row
sar_valid	0/1	Whether valid SAR data exist for this row

Table 5.3: AI Engine Configuration

Engine	Models	Input	Deployment / Status
Crop Engine	RF / XGBoost / RBF-SVM	EVI, NDMI, NDVI, NDWI, SAVI, VV, VH, VV-VH dB	Selected deploy: Random Forest
Phenology Engine	RF / XGBoost / LSTM	LSTM: 11 features × 3 time steps	Selected deploy: LSTM
Stress Engine	Logistic Regression / RF / XGBoost	Optical indices + SAR features	Selected deploy: XGBoost
Irrigation Engine	FAO-style water balance + rules	Weather, crop/stage, Kc, rainfall	No confusion matrix; requires independent validation

Table 5.4: Data and API Endpoint Schema

Method	Endpoint	Purpose
GET	/api/data	Load master metadata, fields and GeoJSON
GET	/api/period/{periodStart}	Load one 8-day period and all field records
GET	/api/field-timeline/{fieldId}	Load complete 19-step field timeline
GET	/api/model-evaluation	Load evaluation metrics and confusion matrices
GET	/api/inference/health	Check persistent Python model readiness
GET	/api/inference/models	List selected deployment models
GET	/api/inference/field/{fieldId}?as_of=YYYY-MM-DD	Return Crop RF, Phenology LSTM and Stress XGBoost predictions
POST	/api/run-pipeline	Rebuild master period data from local inputs

CHAPTER 6

PROJECT IMPLEMENTATION AND RESULT

6.1 Introduction

This chapter documents the implemented Karnal prototype and reports only results supported by the current data artifacts. The system successfully integrates the complete 43,110-row real satellite temporal table, nineteen real 8-day weather files, a 24-band soil stack, model-evaluation artifacts, a seasonal web application and a persistent model-inference service. Results are divided into data-pipeline observations, application implementation and prototype ML evaluation.

6.2 System Implementation

6.2.1 Real Satellite and Weather Integration

The variable-band temporal builder processed 110 satellite TIFFs. Raster layout QA identified 88 full 14-band files and 22 three-band SAR-only files. This confirms that missing optical bands on specific dates are a property of sensor/scene availability rather than file corruption. The pipeline therefore standardizes every record to the same output schema and records missing optical values as NaN.

Table 6.1: Satellite Temporal Coverage by Period

Period Start	Locations	Optical Valid	SAR Valid
2025-06-01	4,311	100.00%	100.00%
2025-06-17	4,311	31.18%	100.00%
2025-07-03	4,311	1.62%	100.00%
2025-07-19	4,311	0.00%	100.00%
2025-08-04	4,311	0.00%	100.00%
2025-08-20	4,311	25.68%	100.00%
2025-09-05	4,311	99.98%	100.00%
2025-09-21	4,311	100.00%	100.00%
2025-10-07	4,311	100.00%	100.00%
2025-10-23	4,311	41.89%	100.00%

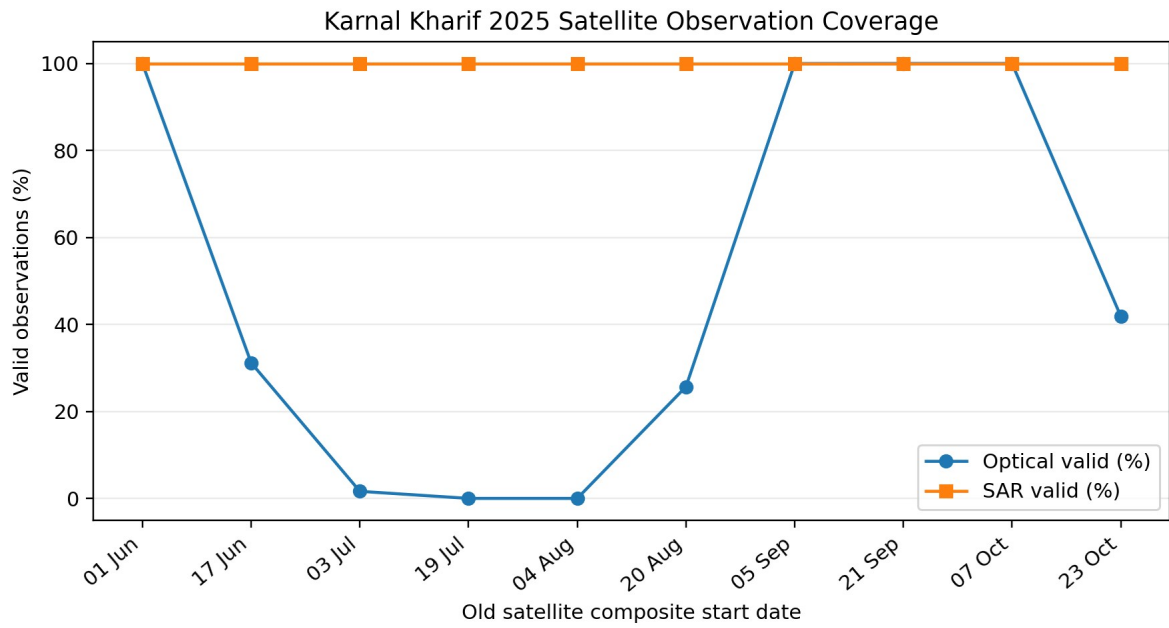


Figure 6.1: Karnal Kharif 2025 Satellite Observation Coverage

The coverage curve explains the value of SAR-optical fusion. On 19 July and 4 August, no valid optical observations are available for the sampled locations, while SAR remains fully available. On 3 July, optical validity is only 1.62%. The application therefore carries the most recent previous optical measurement where appropriate but still uses fresh Sentinel-1 VV/VH from the current satellite period.

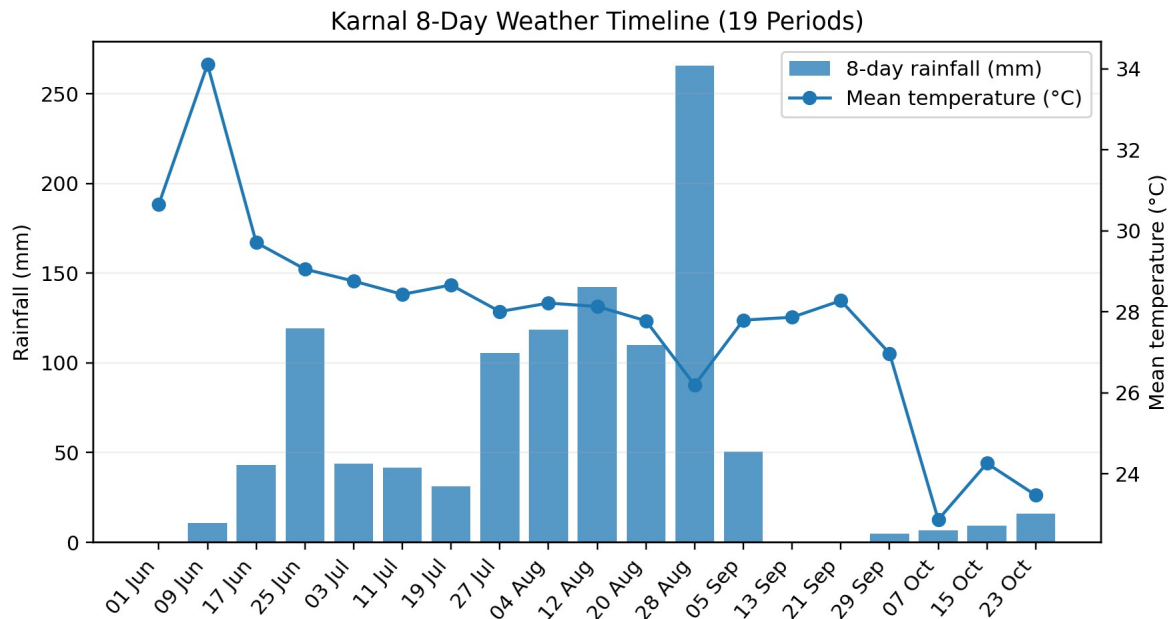


Figure 6.2: 8-Day Weather Timeline

All nineteen weather periods from 1 June through 23 October 2025 are present. The 8-day master timeline uses these files to update rainfall and atmospheric demand twice as frequently as the approximately 16-day satellite schedule. The soil stack remains static and contributes 24 bands across six depths.

6.2.2 Web Application and Role-Specific Interface

The browser application uses Leaflet for map interaction and Chart.js for temporal/summary graphics. A master-period control provides previous/next navigation, a slider, date selection and automatic seasonal playback. The satellite status chip distinguishes fresh, carried, mixed or unavailable

conditions and tracks optical and SAR separately. Role-specific UI filtering reduces information overload and demonstrates how the same analytical backend can support multiple stakeholder workflows.

**PLACEHOLDER - INSERT FINAL AGRICULTURE OFFICER DASHBOARD
SCREENSHOT**

Figure 6.6: Application Screenshot - Agriculture Officer Dashboard

**PLACEHOLDER - INSERT FINAL IRRIGATION OFFICER DASHBOARD
SCREENSHOT**

Figure 6.7: Application Screenshot - Irrigation Officer Dashboard

PLACEHOLDER - INSERT FINAL FARMER VIEW SCREENSHOT

Figure 6.8: Application Screenshot - Farmer View



PLACEHOLDER - INSERT FINAL AI MODEL EVALUATION SCREENSHOT

Figure 6.9: Application Screenshot - AI Model Evaluation

6.2.3 Persistent AI Inference

The deployment package uses a local FastAPI service on a separate Python process. The service loads `crop_engine__random_forest.joblib`, `stress_engine__xgboost.joblib`, `phenology_lstm.keras` and their encoders/preprocessing artifacts once at startup. Field-level inference queries reconstruct the latest valid optical and SAR feature state as of the selected 8-day master date; the LSTM additionally consumes the causal field history up to that date. This replaces hard-coded display labels with actual serialized-model predictions while retaining the warning that the models were trained on prototype labels.

6.3 Performance Evaluation

Table 6.2: Prototype Model Performance Summary

Engine	Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Evaluation Status
Crop Engine	Random Forest	100.00%	1.000	1.000	1.000	Prototype evaluation on derived/pseudo labels
Crop Engine	XGBoost	100.00%	1.000	1.000	1.000	Prototype evaluation on derived/pseudo labels
Crop Engine	SVM RBF	99.65%	0.965	0.998	0.981	Prototype evaluation on derived/pseudo labels
Phenology Engine	Random Forest	100.00%	1.000	1.000	1.000	Prototype evaluation on derived/pseudo labels
Phenology Engine	XGBoost	99.77%	0.997	0.997	0.997	Prototype evaluation on derived/pseudo labels
Phenology Engine	LSTM	80.89%	0.740	0.748	0.740	Prototype field-group temporal evaluation on derived/pseudo NDVI labels
Stress Engine	Logistic Regression	98.96%	0.666	0.895	0.733	Prototype evaluation on derived/pseudo labels
Stress Engine	Random Forest	99.77%	0.999	0.800	0.874	Prototype evaluation on derived/pseudo labels
Stress Engine	XGBoost	100.00%	1.000	1.000	1.000	Prototype evaluation on derived/pseudo labels

The tabular Crop and Phenology baselines achieve extremely high prototype scores because their targets are derived from the same limited feature/label process and are not independent field observations. These numbers therefore should not be interpreted as expected real-world agronomic accuracy. The temporal LSTM is a more demanding test because it uses grouped field splits and five trajectory-derived stage classes; it achieves 80.89% accuracy and a macro-F1 of 0.740 on 647 held-out fields.

The Stress Engine results are also affected by severe label imbalance: the current dataset contains very few Moderate Stress examples compared with Severe Stress. Accuracy can therefore appear high even when minority-class behaviour is weak. Macro-F1 and the confusion matrix are included to make this imbalance visible.

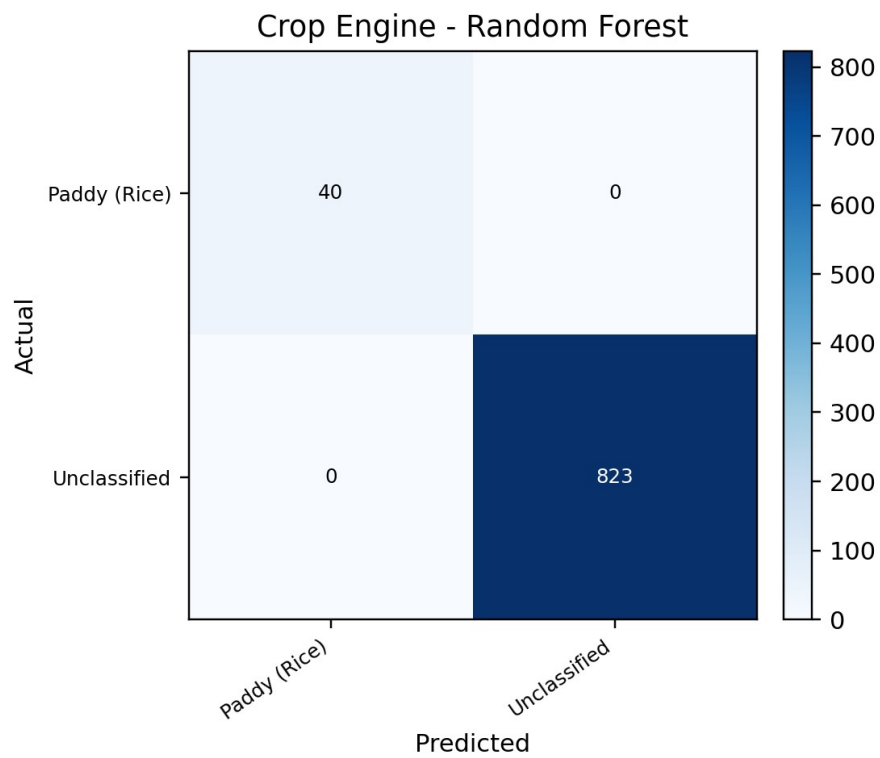


Figure 6.3: Crop Engine - Random Forest Confusion Matrix

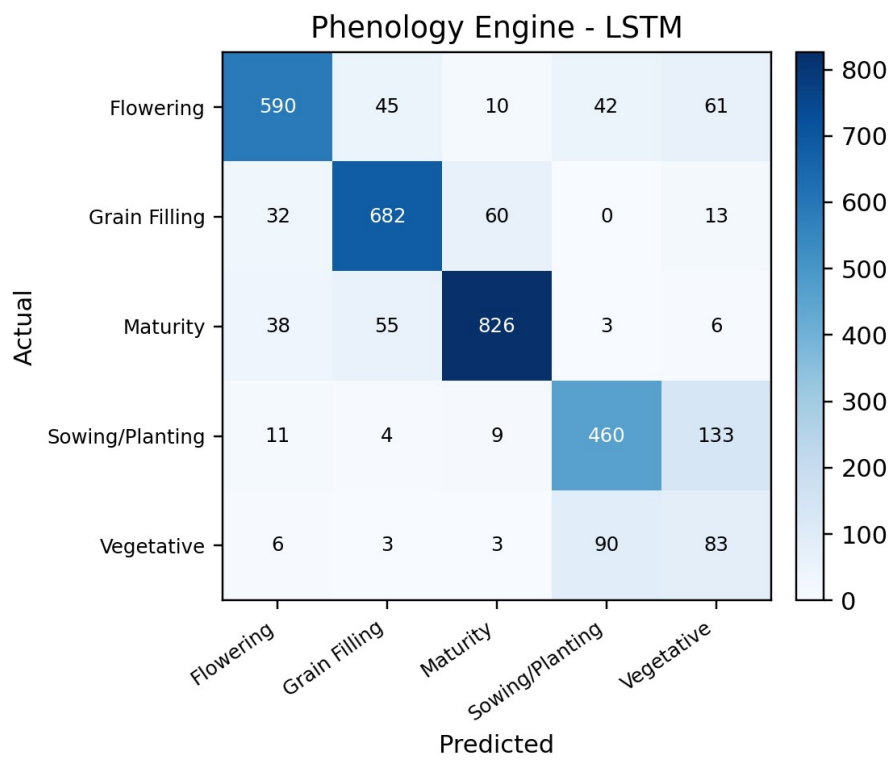


Figure 6.4: Phenology Engine - LSTM Confusion Matrix

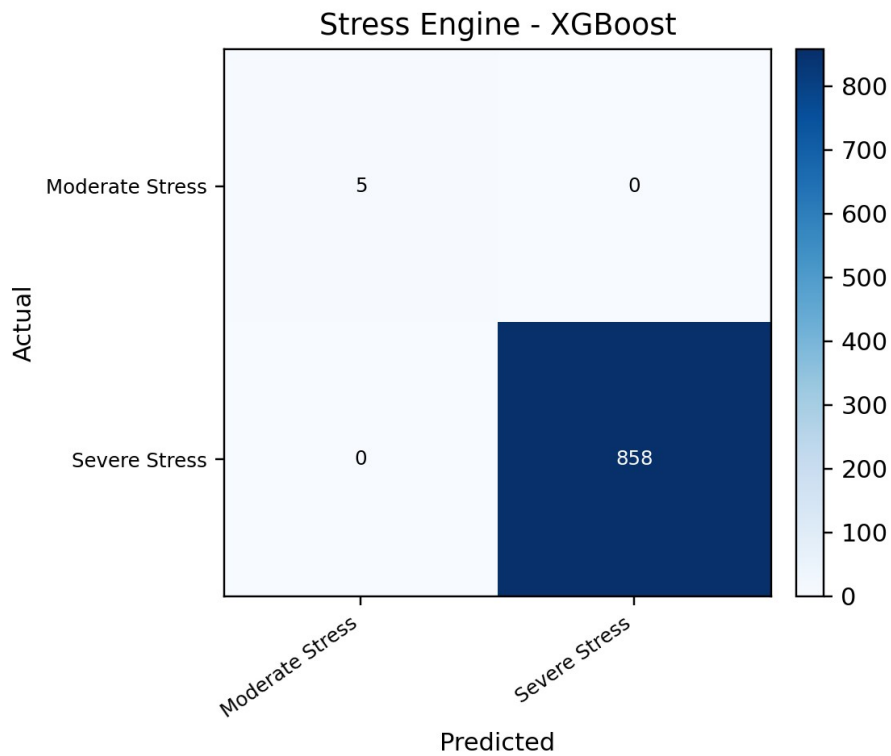


Figure 6.5: Stress Engine - XGBoost Confusion Matrix

Table 6.3: Result Interpretation and Validation Status

Component	Current Result	Interpretation
Satellite data	43,110 real temporal rows; 100% SAR on all 10 dates	Real observation / QA result
Weather data	19 of 19 required 8-day periods available	Real observation / QA result
Soil data	24 bands at six depths	Real static ancillary data
Crop RF	100% prototype accuracy on two derived classes	Implementation check only; not independent crop validation
Phenology LSTM	80.89% accuracy; macro-F1 0.740; 647 held-out fields	Temporal prototype evaluation on NDVI-derived pseudo stages
Stress XGBoost	100% prototype accuracy on highly imbalanced derived labels	Not evidence of independent stress-detection accuracy
Irrigation advisory	8-day FAO-style deficit and priority generated	Prototype rule; no confusion matrix or field validation yet
Application performance	Interactive local demo and persistent model loading implemented	Formal latency/concurrency benchmarking not yet completed

The most defensible result of the current project is the completed, provenance-aware data and application pipeline. It correctly handles sensor gaps, retains real acquisition dates, combines 8-day weather with slower satellite observations, produces live model inference, and exposes validation caveats to users. Independent ground-truth collection is the primary requirement before the model scores can be converted into publishable agronomic performance claims.

CHAPTER 7

CONCLUSION & FUTURE SCOPE

7.1 Conclusion

AgriSense demonstrates an integrated approach to crop and irrigation intelligence for Karnal using real multi-source geospatial data and AI. The implemented system converts a difficult, irregular sensor archive into a consistent seasonal representation while preserving the provenance of every source. A 43,110-row real satellite temporal table, nineteen 8-day weather periods, and a 24-band soil stack are brought together in a 19-step master timeline. Optical and SAR are treated as independent observation streams, allowing the platform to remain useful during periods in which optical coverage is almost absent.

The project implements four decision engines: crop classification, temporal phenology, moisture-stress classification and irrigation advisory. Random Forest, XGBoost and SVM are compared for crop classification; RF, XGBoost and LSTM are compared for phenology; and Logistic Regression, RF and XGBoost are compared for stress. The selected deployment models are executed through a persistent Python service rather than hard-coded in the browser. The irrigation engine uses an interpretable FAO-style water balance instead of an unvalidated black-box classifier.

A major design contribution is the explicit separation of real observations, derived variables, pseudo labels and prototype advisories. Although several model evaluations show high numerical accuracy, the report does not claim these as independent agronomic validation because the current targets are derived and imbalanced. The present achievement is therefore a technically complete prototype and research pipeline that is ready for the next stage: field-labelled validation.

7.2 Limitations

- Independent ground-truth crop, phenology and moisture-stress labels are not yet available for the 4,311 Karnal locations. Current model labels are derived/pseudo labels.
- The observation units are fixed sample points rather than verified cadastral farm polygons; Farmer View therefore uses demo-assigned points rather than actual ownership boundaries.
- Optical coverage is extremely low or zero on several monsoon dates. Carry-forward preserves continuity but does not replace the information content of a fresh optical observation.
- Weather data are coarse relative to individual fields (approximately 5.5 km exported grid), and soil data are 250 m. These resolutions are appropriate as context but should not be interpreted as within-field measurements.
- The irrigation engine uses generic/prototype Kc and effective-rainfall assumptions and has not yet been validated against observed irrigation deliveries, soil moisture or crop water-use measurements.
- Stress labels are highly imbalanced, making simple accuracy misleading; independent stress measurements are required before operational use.
- Role filtering in the current demonstration is presentation-level. Secure production deployment requires authentication, authorization and server-side data filtering.
- Only one Kharif season is included. Inter-annual and multi-season generalization has not yet been established.

7.3 Future Scope

- Collect independent field labels for crop type, agronomic phenological stage, soil moisture, irrigation events and crop condition. These labels should be spatially separated from training data and used for true external validation.
- Replace point samples with verified field boundaries and compute parcel-level temporal statistics. Use spatial/group cross-validation to prevent neighbouring or repeated field observations from leaking across folds.

- Expand to multiple Kharif and Rabi seasons and evaluate transfer across years, crops and neighbouring districts in Haryana.
- Integrate higher-frequency cloud-robust features, including Sentinel-1 temporal derivatives and future Sentinel-2/Landsat fusion where scientifically justified, while retaining source provenance.
- Calibrate crop coefficients, root-zone properties and effective-rainfall rules for local crops and management practices. Validate irrigation recommendations against measured soil moisture and water-delivery records.
- Add uncertainty and Area-of-Applicability layers so officers can see where model predictions are outside the training domain.
- Move production storage to PostGIS/object storage and serve Cloud Optimized GeoTIFFs through a tile service for scalable regional deployment.
- Implement authenticated role-based access control for Farmer, Officer, WUA, GIS Analyst, Admin and Policy/Research users.
- Investigate advanced sequence architectures such as Temporal CNNs or attention-based temporal models only after sufficient independent labelled sequences are available.
- Link advisories with canal-command constraints, water availability and scheduling optimization so field demand can be aggregated into operational water-release planning.

CHAPTER 8

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